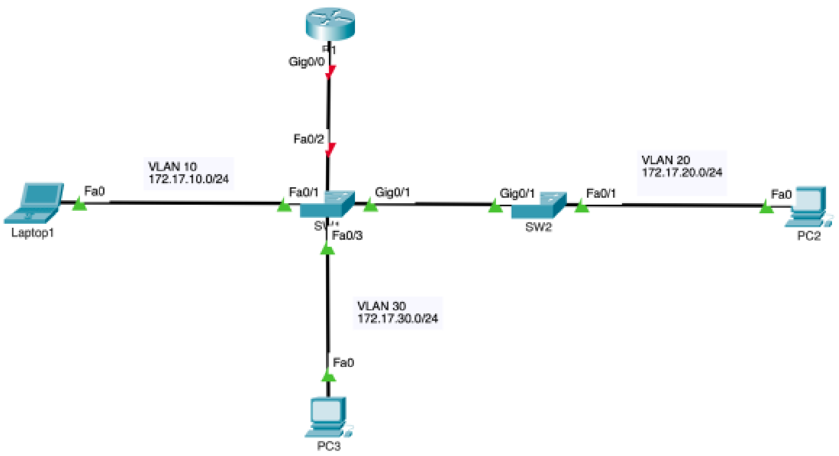
SPENGERGASSE
ausbildung mit zukunft

Name:Hossein Golpayegani	**Labor Protokoll**	Datum: 31.03.2025 Übung Nr.: 3
Klasse: 6ACIF Gruppe:		Raum Nr.: Platz Nr.:
Lehrer: Prof.Spanring, Prof. Horny	Abgabedatum: 07.04.2025	Bewertung:

VLANs mit Inter-VLAN-Routing über Router-on-a-Stick

Teil 1:

- **Aufgabenstellung**



Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	?	?	255.255.255.0	N/A
PC2	NIC	172.17.20.10	255.255.255.0	?
PC3	NIC	172.17.30.10	255.255.255.0	?

Objectives

Part 1: Add VLANs to Switches

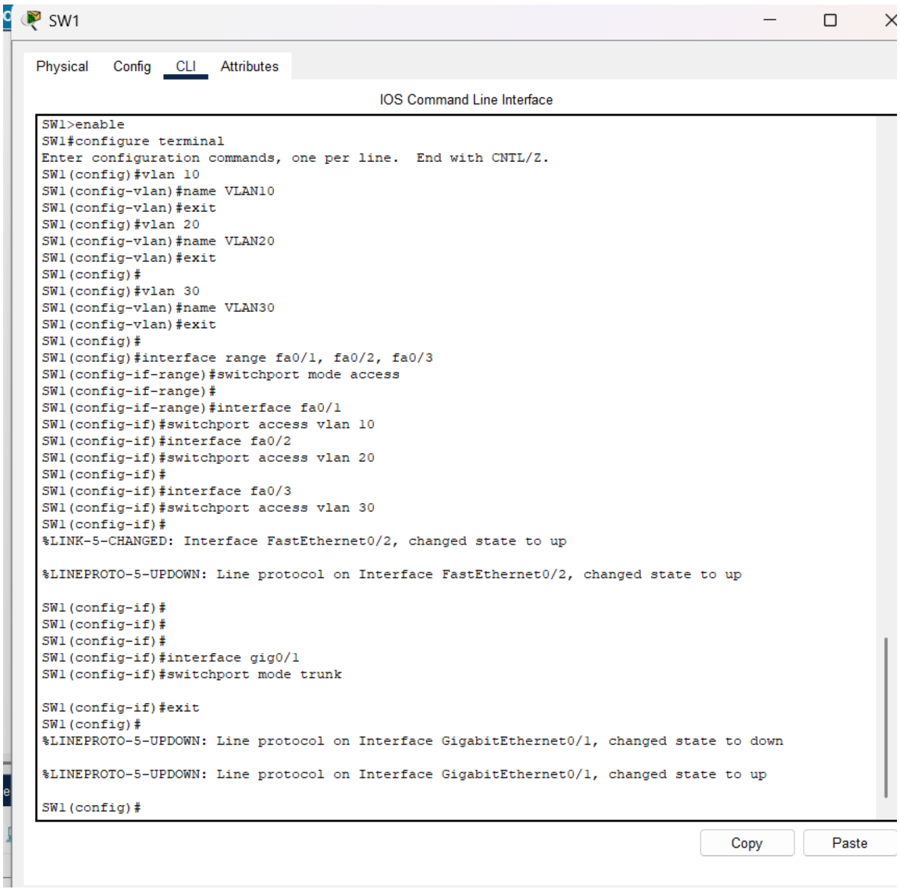
Part 2: Configure Subinterfaces

Part 3: Test Connectivity with Inter-VLAN Routing

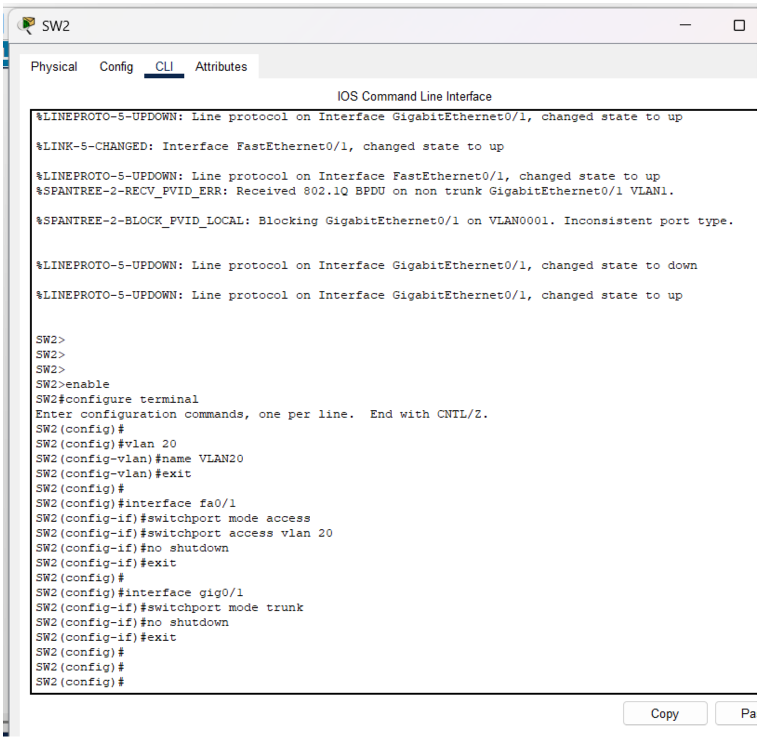
Ziel

- **VLANs einrichten (VLAN 10, 20, 30)**
- **Endgeräte in die VLANs einteilen**
- **Router-on-a-Stick für Inter-VLAN-Routing konfigurieren**
- **Kommunikation zwischen den VLANs ermöglichen**

1. VLANs auf den Switches anlegen
Ports konfigurieren (Access- und Trunk-Ports)

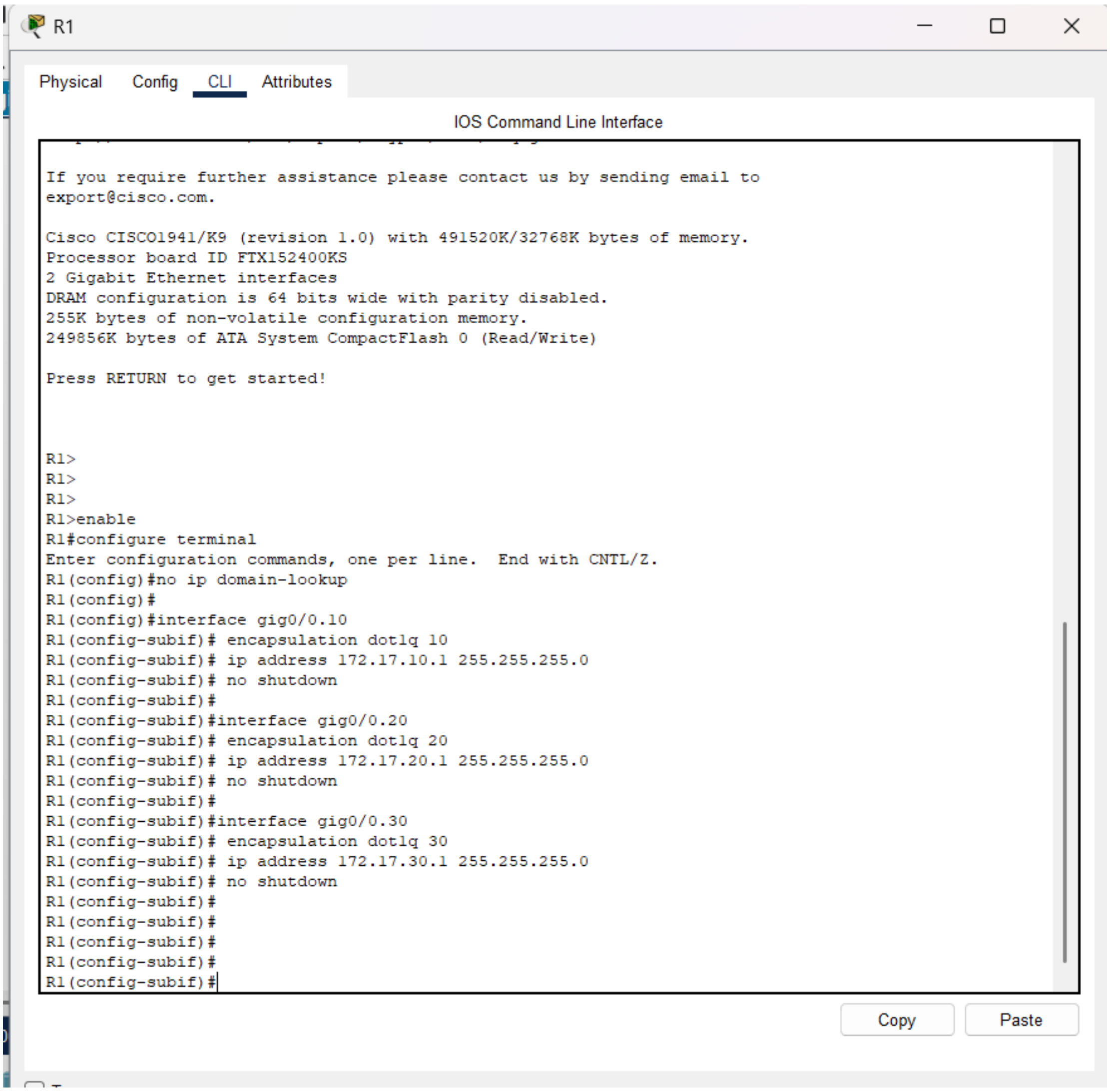



```
SW1>enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#vlan 10
SW1(config-vlan)#name VLAN10
SW1(config-vlan)#exit
SW1(config)#vlan 20
SW1(config-vlan)#name VLAN20
SW1(config-vlan)#exit
SW1(config)#
SW1(config)#vlan 30
SW1(config-vlan)#name VLAN30
SW1(config-vlan)#exit
SW1(config)#
SW1(config)#interface range fa0/1, fa0/2, fa0/3
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#
SW1(config-if-range)#interface fa0/1
SW1(config-if)#switchport access vlan 10
SW1(config-if)#interface fa0/2
SW1(config-if)#switchport access vlan 20
SW1(config-if)#
SW1(config-if)#interface fa0/3
SW1(config-if)#switchport access vlan 30
SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

SW1(config-if)#
SW1(config-if)#
SW1(config-if)#
SW1(config-if)#interface gig0/1
SW1(config-if)#switchport mode trunk

SW1(config-if)#exit
SW1(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

SW1(config)#
```

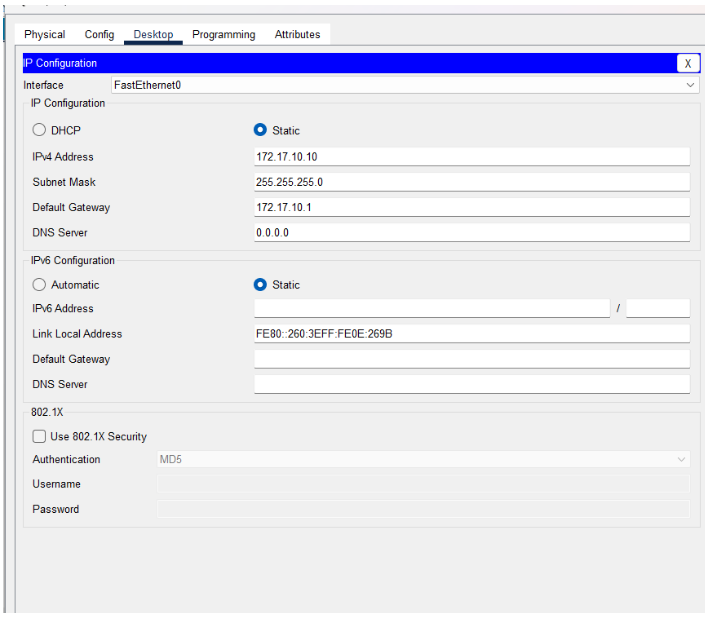


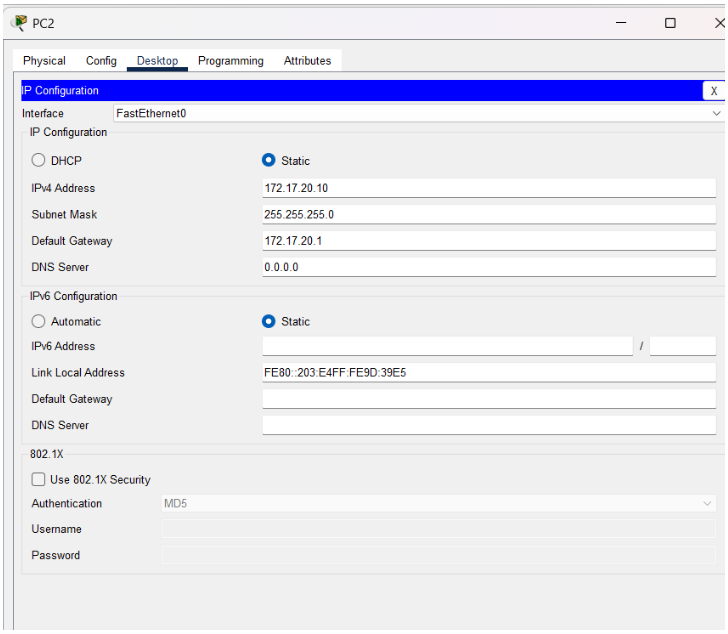

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk GigabitEthernet0/1 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/1 on VLAN0001. Inconsistent port type.


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up


SW2>
SW2>
SW2>
SW2>enable
SW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#
SW2(config)#vlan 20
SW2(config-vlan)#name VLAN20
SW2(config-vlan)#exit
SW2(config)#
SW2(config)#interface fa0/1
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 20
SW2(config-if)#no shutdown
SW2(config-if)#exit
SW2(config)#
SW2(config)#interface gig0/1
SW2(config-if)#switchport mode trunk
SW2(config-if)#no shutdown
SW2(config-if)#exit
SW2(config)#
SW2(config)#
SW2(config)#
```

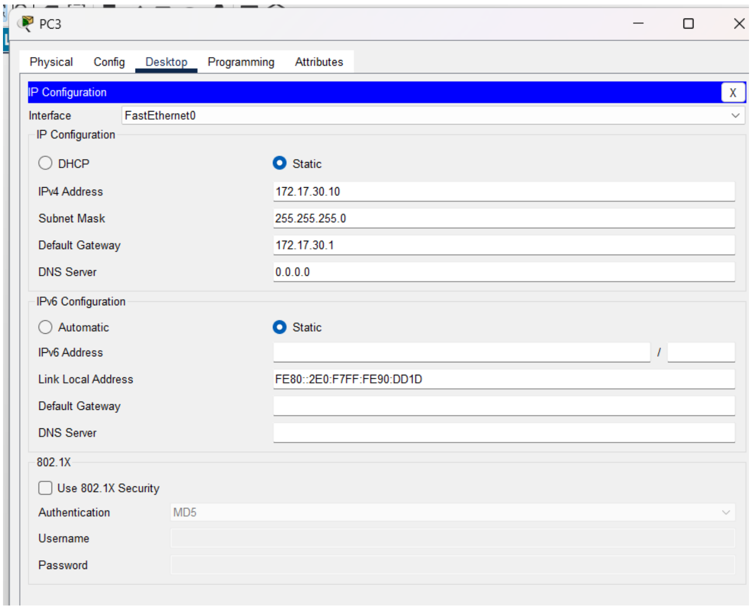
2. Router (R1) – Subinterfaces für Router-on-a-Stick konfigurieren



3. IP-Konfiguration der Endgerät




PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.17.20.10
Subnet Mask
255.255.255.0
Default Gateway
172.17.20.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::203:E4FF:FE9D:39E5
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.17.30.10
Subnet Mask
255.255.255.0
Default Gateway
172.17.30.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2E0:F7FF:FE90:DD1D
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up


SW1>
SW1>
SW1>
SW1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/2
10   VLAN10                           active    Fa0/1
30   VLAN30                           active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,30

SW1>
```

Copy F

```
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  172.17.10.1     YES manual up                    up
GigabitEthernet0/0.20  172.17.20.1     YES manual up                    up
GigabitEthernet0/0.30  172.17.30.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
R1#
```

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```
SW1>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,30

SW1>
```

SW2

Physical Config CLI Attributes

IOS Command Line Interface

```
SW2>
SW2>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/2
20   VLAN20                           active    Fa0/1
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW2>
SW2>
SW2>enable
SW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#interface gig0/1
SW2(config-if)#switchport mode trunk
SW2(config-if)#no shutdown
SW2(config-if)#exit
SW2(config)#
SW2(config)#
SW2(config)#
SW2(config)#
```

Copy Paste

```
SW1#
SW1#
SW1#
SW1#
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)#switchport mode trunk

SW1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#
```

Copy

4. Test der Konnektivität (Ping)
Alle Geräte konnten erfolgreich gegeneinander pingen

Laptop1

```
C:\>ping 172.17.30.10

Pinging 172.17.30.10 with 32 bytes of data:

Reply from 172.17.30.10: bytes=32 time<1ms TTL=127
Reply from 172.17.30.10: bytes=32 time<1ms TTL=127
Reply from 172.17.30.10: bytes=32 time<1ms TTL=127
Reply from 172.17.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 172.17.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Top

```
C:\>ping 172.17.10.10

Pinging 172.17.10.10 with 32 bytes of data:

Reply from 172.17.10.10: bytes=32 time<1ms TTL=128
Reply from 172.17.10.10: bytes=32 time<1ms TTL=128
Reply from 172.17.10.10: bytes=32 time=11ms TTL=128
Reply from 172.17.10.10: bytes=32 time=2ms TTL=128

Ping statistics for 172.17.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>
```

Command Prompt

```
Request timed out.
Reply from 172.17.20.10: bytes=32 time<1ms TTL=127
Reply from 172.17.20.10: bytes=32 time<1ms TTL=127
Reply from 172.17.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 172.17.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.17.20.10

Pinging 172.17.20.10 with 32 bytes of data:

Reply from 172.17.20.10: bytes=32 time=1ms TTL=127
Reply from 172.17.20.10: bytes=32 time<1ms TTL=127
Reply from 172.17.20.10: bytes=32 time<1ms TTL=127
Reply from 172.17.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 172.17.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

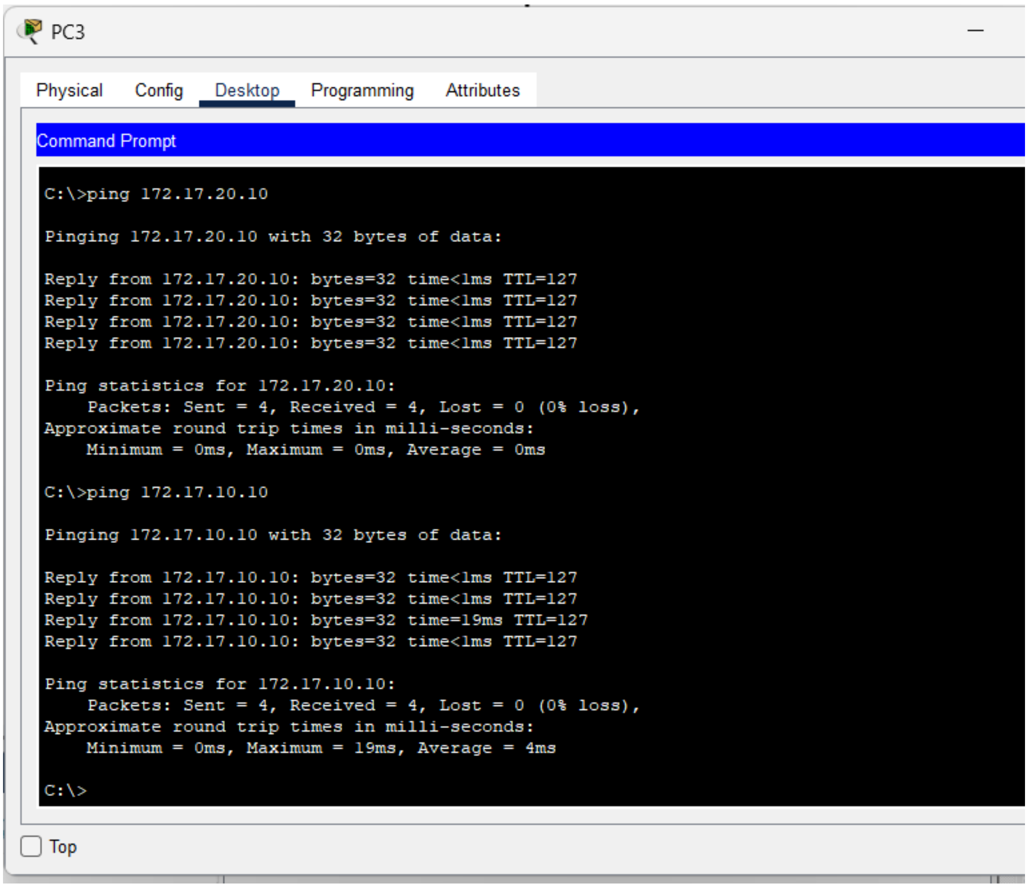
PC2

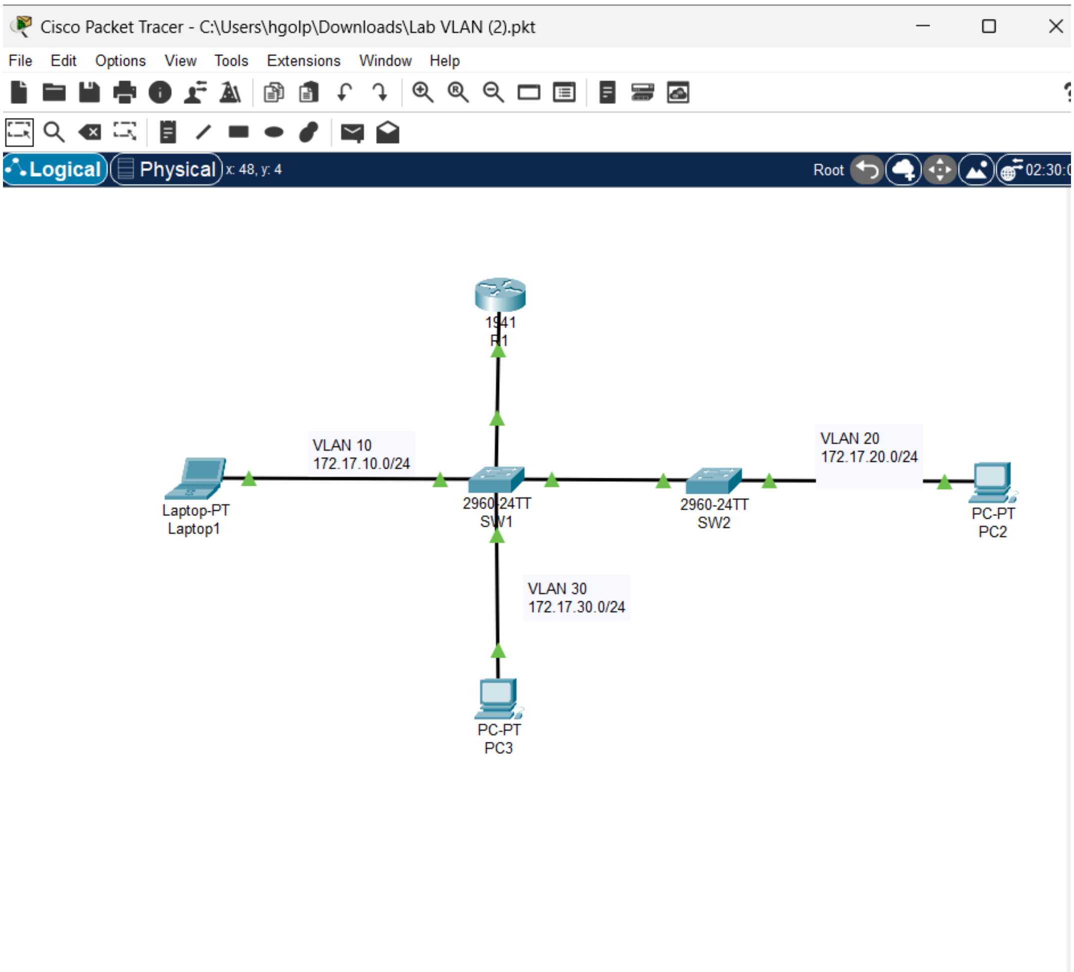
Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 172.17.30.10

Pinging 172.17.30.10 with 32 bytes of data:

Reply from 172.17.30.10: bytes=32 time=1ms TTL=127
Reply from 172.17.30.10: bytes=32 time<1ms TTL=127
Reply from 172.17.30.10: bytes=32 time<1ms TTL=127
Reply from 172.17.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 172.17.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.17.10.10

Pinging 172.17.10.10 with 32 bytes of data:

Reply from 172.17.10.10: bytes=32 time=7ms TTL=127
Reply from 172.17.10.10: bytes=32 time<1ms TTL=127
Reply from 172.17.10.10: bytes=32 time=1ms TTL=127
Reply from 172.17.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 172.17.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 2ms

C:\>
```

Top


PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 172.17.20.10
Pinging 172.17.20.10 with 32 bytes of data:
Reply from 172.17.20.10: bytes=32 time<1ms TTL=127
Reply from 172.17.20.10: bytes=32 time<1ms TTL=127
Reply from 172.17.20.10: bytes=32 time<1ms TTL=127
Reply from 172.17.20.10: bytes=32 time<1ms TTL=127
Ping statistics for 172.17.20.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 172.17.10.10
Pinging 172.17.10.10 with 32 bytes of data:
Reply from 172.17.10.10: bytes=32 time<1ms TTL=127
Reply from 172.17.10.10: bytes=32 time<1ms TTL=127
Reply from 172.17.10.10: bytes=32 time=19ms TTL=127
Reply from 172.17.10.10: bytes=32 time<1ms TTL=127
Ping statistics for 172.17.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 19ms, Average = 4ms
C:\>
Top


Cisco Packet Tracer - C:\Users\hgolp\Downloads\Lab VLAN (2).pkt
File
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Physical
x: 48, y: 4
Root
1941
R1
VLAN 10
172.17.10.0/24
Laptop-PT
Laptop1
2960-24TT
SW1
2960-24TT
SW2
VLAN 20
172.17.20.0/24
PC-PT
PC2
VLAN 30
172.17.30.0/24
PC-PT
PC3